Math Exploration

Application of Calculus and Trigonometry in Designing the Winding Mountain Road

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1 Introduction

1.1 Rationale

During the Chinese New Year, I traveled back to my hometown, Chongqing, with my family. When driving between mountains and forests, I figured out the road from the bottom of the mountain to its top was not a straight line. The road was actually winding around the body of the mountain. However, in some specific places when the road is a little steeper, some cars were unable to climb up. This interests me since the construction of the road needs to not only enable people to get to the top, but also fulfills the ability to climb of the car. Hence, in order to fulfill such two requirements and also minimize the construction cost, I start an exploration based on the situation I saw back my hometown.

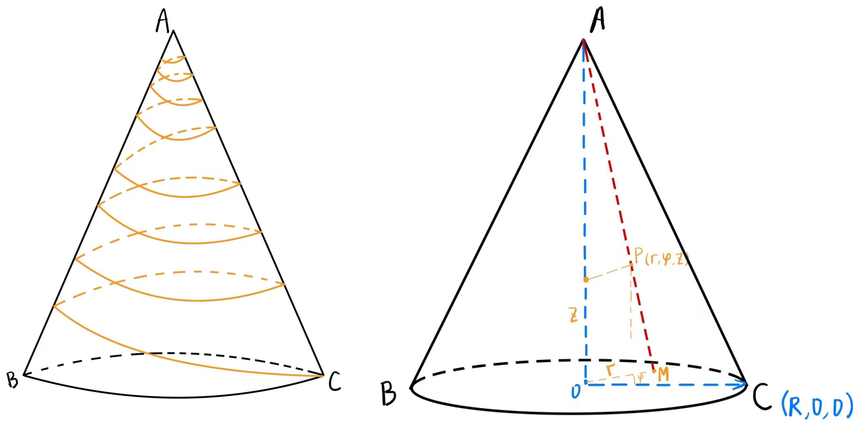
1.2 Aim

This exploration aims to **calculate the relationship between the length of mountain roads with the maximum slope a car can climb**. In order to achieve this aim, I plan to build a cylindrical coordinate and model the mountain as (part of) a circular cone to do my calculations. These may apply my knowledge from different mathematics areas for instance: Calculus, Trigonometry, Geometry and etc.

1.3 Research Plan - Methodology

Firstly, I model the mountain as an ideal circular cone. The top vertex is labeled A, and two ending points of a base diameter are denoted as B and C.

Next, I build a cylindrical coordinate [1] based on Figure 1a and is shown below on Figure 1b. [AR] is a segment perpendicular to the sub-face of the cone and point R is its foot of perpendicular. Therefore, point C can be named as (R,0,0) since [RC] is the radius of sub-face which has a length of R. The second component represents the angle $\angle P'$OC where P' is the projection of point P on the sub-face. The third component represents the distance between the target point and the sub-face.



(a) (b)

Figure 1: Schematic diagram of the winding mountain road

In this model, r means the distance between the projection of the position of the car on the sub-face and the center of the sub-face. φ means the angular displacement between r and the horizontal line on the sub-face. z is the altitude of the car.

Generally, I will find a point on Figure 1b, make P as an example. By using the method describe above, let its coordinates be (r,φ,z). I will make C as the starting point of the winding mountain road and [PC] to be the length of the road. After that, I will express the slope of such road by using trigonometry to match with the ability of cars.

Finally, I will set up equations between the winding mountain road and its slope and if possibly improve the model to make it closer to mountains in the real world to check the universality of my calculations.

2 Main Body of Exploration

2.1 Modeling and calculating the slope of the mountain

Generally, a slope of an inclined surface is calculated by dividing its vertical height to its horizontal length. In right triangle △ABC, ∠A is 90°. Therefore, the slope of [BC] can be illustrated by:

$$m_{[BC]} = \frac{[AC]}{[AB]}$$

Similarly, the slope of the cylinder-mountain in my exploration can also be calculated by using this method.

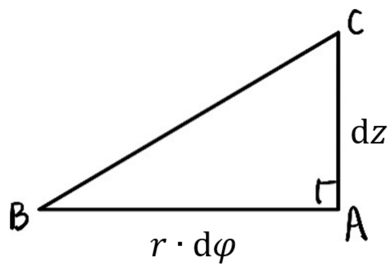


Figure 2: The slope of two close points at different level

In Figure 1b, the sub-face of the mountain is a circle, therefore, if a car wants to climb up around the mountain, it's path must also be a winding path. As a result, for each small part of the winding mountain road, the horizontal distance it moved is an arc length of a circle. By using cylindrical coordinate, the arc length for every angle φ moved can be illustrated as $r \cdot \mathrm{d}\varphi$. Hence, the slope of the mountain road can be illustrated as:

$$m = \frac{\text{changeinverticaldistance}}{\text{changeinhorizontaldistance}} = \frac{\mathrm{d}z}{r \cdot \mathrm{d}\varphi}$$

In addition, consider to build the shortest mountain road, the slope of the road should exactly be the maximum slope a car can climb (labeled α). Therefore,

$$\tan \alpha = m = \frac{\mathrm{d}z}{r \cdot \mathrm{d}\varphi}$$

2.2 Conical mountain

Making l the length of the mountain road, α the maximum slope a car can climb.

As modeling by Figure 3, P is the position of the car. E is the projection of the car on the vertical axis, and H is the distance between mountain top and E.

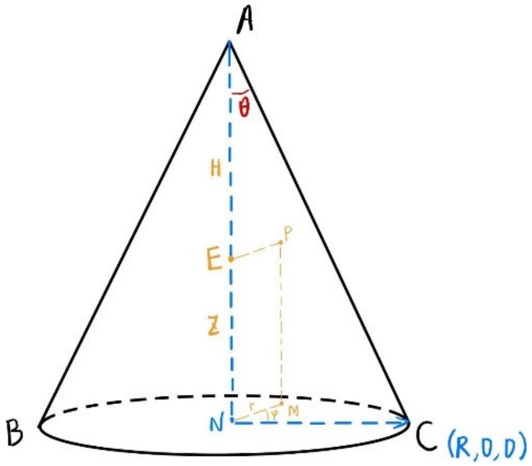


Figure 3: Schematic diagram of a car on the mountain

θ is the angle between the vertical axis and the slant height. Therefore,

$$\tan\theta = \frac{r}{H-z}$$

Hence,

$$r = \tan\theta \cdot (H-z)$$

$$\because \tan\alpha = \frac{\mathrm{d}z}{r\cdot \mathrm{d}\varphi}$$

$$\therefore \tan\alpha \cdot \tan\theta \cdot \mathrm{d}\varphi = \frac{\mathrm{d}z}{H-z}$$

By separate the variables:

$$\int \tan\alpha \cdot \tan\theta \cdot \mathrm{d}\varphi = \int \frac{1}{H-z}\cdot \mathrm{d}z$$

$$(\tan\alpha \cdot \tan\theta)\cdot\varphi = -\ln|H-z| + c$$

When climbing up a mountain, we usually start from the bottom where the altitude is 0 and the angular displacement against the horizontal is also 0. Hence, set an initial condition where φ=0 and z=0.

$$\therefore c = \ln H$$

$$\begin{aligned}\tan\alpha \cdot \tan\theta \cdot \varphi &= -\ln(H-z) + \ln H \\ &= \ln \frac{H}{H-z}\end{aligned}$$

$$e^{\tan\alpha \cdot \tan\theta \cdot \varphi} = \frac{H}{H-z} \tag{1}$$

As it has been mentioned above $\tan\theta = \frac{r}{H-z}$. Hence,

$$\frac{1}{H-z} = \frac{\tan\theta}{r}$$

Substitute the equation above to the right hand side of equation 1:

$$\begin{aligned}\frac{\tan\theta}{r} \cdot H &= e^{\tan\alpha \cdot \tan\theta \cdot \varphi} \\ r &= H \cdot \tan\theta \cdot e^{-\tan\alpha \cdot \tan\theta \cdot \varphi}\end{aligned}$$

Consider Figure 2 (*The slope of two close points at different level*) again, [BC] can be used to illustrate the length of a small part of the road (dl). This can be illustrated as:

$$\mathrm{d}l = \frac{r \cdot \mathrm{d}\varphi}{\cos\alpha} = r \cdot \mathrm{d}\varphi \cdot \sec\alpha$$

Therefore,

$$\begin{aligned}\mathrm{d}l &= H \cdot \tan\theta \cdot e^{-\tan\alpha \cdot \tan\theta \cdot \varphi} \cdot \sec\alpha \cdot \mathrm{d}\varphi \\ \int \mathrm{d}l &= \int H \cdot \tan\theta \cdot e^{-\tan\alpha \cdot \tan\theta \cdot \varphi} \cdot \sec\alpha \cdot \mathrm{d}\varphi\end{aligned}$$

As the car is climbing up, the radius of the horizontal cross section will decrease. Therefore, the level raised every round will become very small. Hence, the car should engage in several rounds to reach the top. As a result, the upper limit for this integral is ∞.

Hence,

$$l = \int_0^\infty H \cdot \tan\theta \cdot e^{-\tan\alpha \cdot \tan\theta \cdot \varphi} \cdot \sec\alpha \cdot \mathrm{d}\varphi$$

In this case, the upper limit of this integration is ∞. Hence, improper integration can be used to solve such type of integration:

$$
\begin{aligned}
l &= \lim_{b\to\infty} \int_0^b H \cdot \tan\theta \cdot e^{-\tan\alpha \cdot \tan\theta \cdot \varphi} \cdot \sec\alpha \cdot \mathrm{d}\varphi \\
&= \lim_{b\to\infty} [-H \cdot \csc\alpha \cdot e^{-\tan\alpha \cdot \tan\theta \cdot \varphi}]_0^b \\
&= -H \cdot \csc\alpha \cdot e^{-(\tan\alpha \cdot \tan\theta)\cdot b} - (-H \cdot \csc\alpha \cdot e^{-(\tan\alpha \cdot \tan\theta)\cdot 0}) \\
&= -H \cdot \csc\alpha \cdot e^{\infty} + H \cdot \csc\alpha \cdot e^{0}
\end{aligned}
$$

$$\therefore l = H \cdot \csc\alpha$$

2.2.1 Reflection

From the result of 2.2, it can be noticed that there is a linear relationship between the length of mountain road (l) and $\csc\alpha$ with a gradient value of H. In other words, the length of the winding mountain road has no relationship to θ, i.e., the length of the road is not relevant to the shape of the mountain. Therefore, I decide to do some further explorations on a mountain that has a platform on the midway which served as the destination of the mountain road.

2.3 Conical mountain with platform on the top

This condition is illustrated by Figure 4 below:

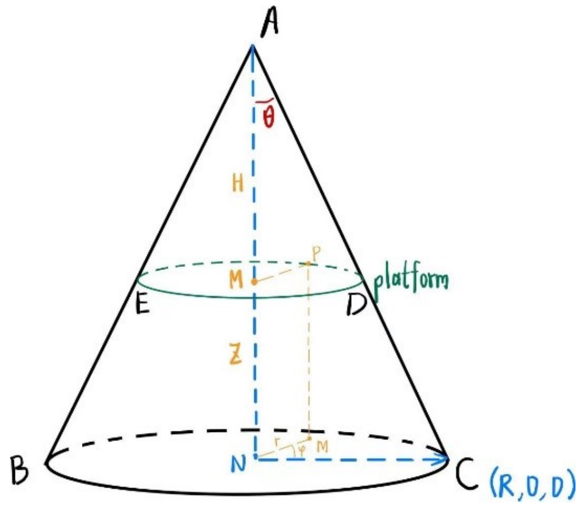


Figure 4: A mountain with a platform on its midway

The green ellipse on Figure 4 is the platform on the cylinder-mountain. In order to simplify the model, I extract the part of the mountain which only the road will be built around. It is illustrated by Figure 5 below:

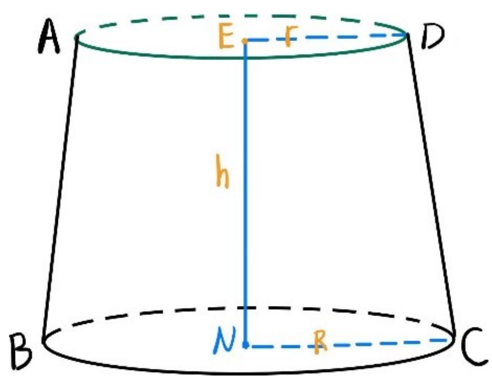


Figure 5: Mountain that the roads will be built around

In Figure 5, the upper circle illustrates the platform, which has a radius r. The sub-face of this bucket is a circle with a radius R. [EN] represents the same [MN] in Figure 4. However, the length of [EN] in Figure 5 is h instead of z. Similarly, the equation that is used in 2.2 can also be used here.

$$\tan\alpha \cdot \tan\theta \cdot \varphi = -\ln(H - z) + \ln H$$

Replace z with h:

$$\varphi = \frac{1}{\tan\alpha \cdot \tan\theta} \cdot \ln(\frac{H}{H-h})$$

Through Figure 4, △AED ∼ △ABC, hence,

$$\frac{r}{H-h} = \frac{R}{H}$$
$$H = \frac{R \cdot h}{R-r}$$

Hence,

$$\tan\theta = \frac{R}{H} = \frac{R-r}{h}$$

Therefore, the angle of the road that projected on sub-face (φ_h) can be illustrated:

$$\varphi_h = \frac{1}{\tan\alpha \cdot \frac{R-r}{h}} \cdot \ln\frac{H}{H-h}$$
$$\because \frac{H}{H-h} = \frac{R}{r}$$
$$\therefore \varphi_h = \frac{h}{\tan\alpha \cdot (R-r)} \cdot \ln\frac{R}{r}$$

Therefore, the length of the road l can be calculated:

$$\begin{aligned} l &= \int_0^{\varphi_h} H \cdot \tan\theta \cdot e^{-\tan\alpha \cdot \tan\theta \cdot \varphi} \cdot \sec\alpha \cdot \mathrm{d}\varphi \\ &= [-H \cdot \csc\alpha \cdot e^{-\tan\alpha \cdot \tan\theta \cdot \varphi}]_0^{\varphi_h} \\ &= H \cdot \csc\alpha - (H-h) \cdot \csc\alpha \end{aligned}$$
$$\therefore l = h \cdot \csc\alpha$$

2.3.1 Reflection

Comparing the result I get for this part, the only difference to result in 2.2 is that the gradient value for this function changes from H to h with no other changes. In other words, there is nearly no changes for the relationship between the length of mountain-road to the maximum slope the car can climb in both situations. Therefore, I plan to do deeper explorations which I mentioned in introduction part which is complicate my model to make it closer to a real mountain, and calculate the relationship under that condition.

2.4 Spherical mountain

In order to make my model real, I make the slope of the mountain decreases as the height increases. The model is drawn below in Figure 6,

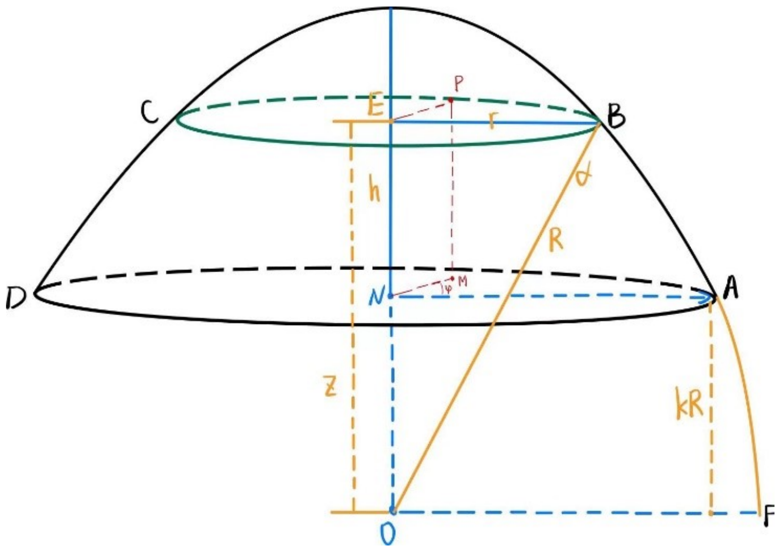


Figure 6: Schematic diagram of a car the mountain closer in reality

In Figure 6, the mountain is modeled closer to a real mountain with a curve-like edge. Meanwhile, kR represents the length of A to the radius of the sphere where the mountain is in. Since the slope of the mountain decreases with increasing in height, therefore, when at height h , the car can climb up the mountain directly from the mountain edge instead of driving on mountain roads. It is considered to be a platform in latter discussions. z represents the length of [EO] which is the distance between the place where the car can directly climb up to the center of the sphere O. In this graph, [EO], [EB] and [OB] form a right triangle, thus by the Pythagorean theorem:

$$z^2 + r^2 = R^2$$

Moreover,

$$\because \tan\alpha = \frac{\mathrm{d}z}{r \cdot \mathrm{d}\varphi}$$
$$\therefore \tan\alpha = \frac{\mathrm{d}z}{\sqrt{R^2 - z^2} \cdot \mathrm{d}\varphi}$$
$$\tan\alpha \cdot \mathrm{d}\varphi = \frac{\mathrm{d}z}{\sqrt{R^2 - z^2}}$$

Using methods of solving differential equations:

$$\int \tan\alpha \cdot \mathrm{d}\varphi = \int \frac{1}{\sqrt{R^2 - z^2}} \cdot \mathrm{d}z \tag{2}$$

Consider the right hand side of the integration: $\int \frac{1}{\sqrt{R^2 - z^2}} \cdot \mathrm{d}z$

Using integration by substitution [2] to solve the integration above, let $z = R \cdot \sin t$.

Therefore,

$$\frac{\mathrm{d}z}{\mathrm{d}t} = R \cdot \cos t$$
$$\mathrm{d}z = R \cdot \cos t \cdot \mathrm{d}t$$

Hence,

$$\begin{aligned} I &= \int \frac{1}{\sqrt{R^2 - (R - \sin t)^2}} \cdot R \cdot \cos t \cdot \mathrm{d}t \\ &= \int \frac{1}{R \cdot \sqrt{1 - (\sin t)^2}} \cdot R \cdot \cos t \cdot \mathrm{d}t \\ &= \int \frac{1}{R \cdot \cos t} \cdot R \cdot \cos t \cdot \mathrm{d}t \\ &= \int \mathrm{d}t \\ &= t + c \end{aligned}$$

As it has been shown above, $z = R \cdot \sin t$. Therefore,

$$t = \arcsin \frac{z}{R}$$

As a result, the answer to the integration $\int \frac{1}{\sqrt{R^2 - z^2}} \cdot \mathrm{d}z$ is:

$$I = \arcsin \frac{z}{R} + c$$

Therefore, for the answer to equation 2 is:

$$\tan\alpha \cdot \varphi = \arcsin \frac{z}{R} + c$$

In this case, I still assume that the car climbs from the bottom. However, the car does not start from the radius of the mountain sphere horizontally. Therefore, set an initial condition where φ=0 and z=kR. For z, since the car starts at the foot of the mountain, the value of z will be the same as the length of [NO]=kR.

Therefore,

$$0 = \arcsin k + c$$

$$c = -\arcsin k$$

$$\tan\alpha \cdot \varphi = \arcsin\frac{z}{R} - \arcsin k$$

$$\frac{z}{R} = \sin(\tan\alpha \cdot \varphi + \arcsin k) \tag{3}$$

Consider Figure 6 again,

$$\because \angle EOB = \alpha$$

$$\therefore z = R \cdot \cos\alpha$$

Substitute $z = R \cdot \cos\alpha$ into equation 3 and rearrange it:

$$\varphi \cdot \tan\alpha = \arcsin\frac{R \cdot \cos\alpha}{R} - \arcsin k$$
$$\varphi \cdot \tan\alpha = \frac{\pi}{2} - \alpha - \arcsin k$$

Therefore, φ_h which is φ when the car reaches the platform can be illustrated as:

$$\frac{1}{\tan\alpha} \cdot (\frac{\pi}{2} - \alpha - \arcsin k)$$

Similar in 2.3,

$$\mathrm{d}l = r \cdot \sec\alpha \cdot \mathrm{d}\varphi$$

$$l = \int_0^{\varphi_h} r \cdot \sec\alpha \cdot \mathrm{d}\varphi$$

However, the l here represents the mountain road the car climbs to the platform, therefore, the total distance traveled by the car can be illustrated as:

$$l = \int_0^{\varphi_h} r \cdot \sec\alpha \cdot \mathrm{d}\varphi + \alpha \cdot R$$

Where $\alpha \cdot R$ is the length of arc with a slope that fulfills the maximum slope a car can climb up directly.

Moreover, since $\frac{z}{R} = \sin(\tan\alpha \cdot \varphi + \arcsin k)$,

$$\begin{aligned} z &= R \cdot \sin(\tan\alpha \cdot \varphi + \arcsin k) \\ r &= \sqrt{R^2 - z^2} \\ &= \sqrt{R^2 - R^2 \cdot \sin\left(\tan\alpha \cdot \varphi + \arcsin k\right)^2} \\ &= R \cdot \cos(\tan\alpha \cdot \varphi + \arcsin k) \end{aligned}$$

$$\therefore l = \int_0^{\varphi_h} [R \cdot \cos(\tan\alpha \cdot \varphi + \arcsin k) \cdot \sec\alpha] \cdot \mathrm{d}\varphi + \alpha \cdot R$$

At this point, I figure out that I can make k as an independent variable and $\frac{l}{R}$ as the dependent variable since I can summarize the whole equation making all l and R on one side and k on the other.

$$\begin{aligned} \frac{l}{R} &= \int_0^{\frac{1}{\tan\alpha}\cdot(\frac{\pi}{2}-\alpha-\arcsin k)} [\cos(\tan\alpha \cdot \varphi + \arcsin k) \cdot \sec\alpha] \cdot \mathrm{d}\varphi + \alpha \\ &= [\csc\alpha \cdot \sin(\tan\alpha \cdot \varphi + \arcsin k)]_0^{\frac{1}{\tan\alpha}\cdot(\frac{\pi}{2}-\alpha-\arcsin k)} \\ &= \csc\alpha \cdot \sin(\frac{\pi}{2} - \alpha) - k \cdot \csc\alpha \end{aligned}$$

Therefore,

$$\frac{l}{R} = -k \cdot \csc\alpha + \cot\alpha + \alpha$$

2.4.1 Reflection

In this part, I expand my research to a mountain much closer to the reality successfully. As the result has shown, $\frac{l}{R}$ has a linear relationship with k. The gradient of this function is $-\csc\alpha$ with a y-intercept at $(0, \cot\alpha + \alpha)$. This means that with the increasing in the value of k, there is a decreasing in the value of $\frac{l}{R}$. Moreover, the equation above can also be translated into $l = (R \cdot \cos\alpha - k \cdot R) \cdot \csc\alpha + \alpha \cdot R$. Therefore, the length of the winding mountain road can be considered as a straight slope with an inclining angle α. The road will raise the altitude of the car by $R \cdot \cos\alpha - k \cdot R$. As applied to the model drawn in Figure 7, the winding mountain road will become shorter for a flatter mountain and longer for a steeper one.

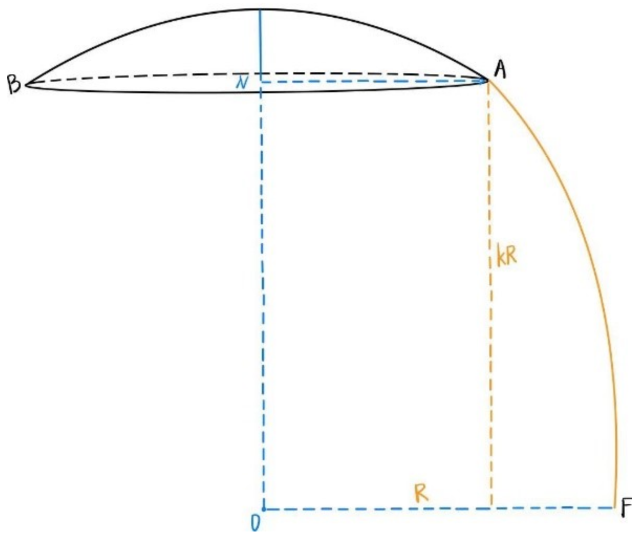


Figure 7: A mountain with a larger kR

3 Conclusion

I am glad that by using Algebra, Geometry, Trigonometry, and Calculus in mathematics, I complete my aim of this exploration: **calculating the relationship between the length of mountain roads with the maximum slope a car can climb**. With a specific maximum slope that a car can climb, the length of the winding mountain road will be longer for a higher mountain which matches with my common sense. Moreover, for a specific mountain, the length of the winding mountain road will decrease for an increase in the maximum slope a car can climb. (Taking domain from 0 to $\frac{\pi}{2}$.) These are illustrated again through the following functions:

$$l = H \cdot \csc\alpha$$

$$l = h \cdot \csc\alpha$$

This is a very interesting result, since both equations show a positive relationship between the length of the winding mountain road and the vertical height of the end point of the car. The coefficient of such relationship is $\csc\alpha$. In other word, $\frac{l}{H} = \frac{1}{\sin\alpha}$, and $\frac{H}{l} = \sin\alpha$. This can be illustrated through Figure 8 below:

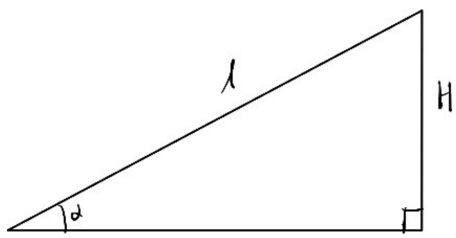


Figure 8: relationship between l, H, and α

From Figure 8, the length of the winding mountain road of an ideal conical mountain can be simply illustrated as a large triangle with a slope equal to the maximum slope a car can climb.

After the conical mountain, I create a better model to simulate the mountain in our real world. This helps me concluded that as long as there is a winding mountain road, its length is always equal to the altitude raised over the sine value of the maximum slope a car can climb. In all these sophisticated equations, all the length of winding mountain roads will increase with the height of mountain and decrease with increase of maximum slope a car can climb. These match my common sense and provide accuracy to my exploration.

References

[1] Wikipedia. (2021). Cylindrical coordinate system. [online] Available at: https://en.wikipedia.org/wiki/Cylindrical_coordinate_system.

[2] Martin, D. (2012). Mathematics for the international student : Mathematics HL (Core). Adelaide: Haese Mathematics.